AMC Movie Planner v0.0.6 - Testing Document & Regression Suite

Code Version: v0.0.6

Focus Areas: Asynchronous Pipeline Performance, Rotten Tomatoes Scraper Resilience (curl_cffi), Legacy & Re-Release Year Candidate Fallbacks, Metacritic TBD State Handling, Title Sanitization Regex, and Unified Embed Architecture.

1. Boundary Value Analysis & Metric Thresholds (BVA)

1.1 Ingestion Performance & Concurrency Thresholds

The asynchronous refresh pipeline utilizes aiohttp for API metadata and curl_cffi for browser TLS spoofing. This test validates latency and throughput boundaries.

Test Path	Parameter / Condition	Expected Behavior
Below Threshold	5 Movies in Schedule (~15 requests)	Cache populates in < 1.5s.
Target Operational	15–20 Movies in Schedule (~45–60 concurrent async requests)	Cache populates in **2.0s – 4.8s** (Target: ~2.63s).
High Load Bound	35+ Movies in Schedule (~100+ concurrent requests)	Cache populates in < 6.5s without hitting rate limits or event loop blocks.
Timeout Exception	External HTTP response >7.0s	Individual request times out gracefully; returns default placeholder metadata without hanging pipeline.

1.2 Title Search & Regex Sanitization (clean_title_for_search)

Validates edge cases in stripping format tags, event qualifiers, and trailing punctuation before dispatching search queries to TMDB, OMDb, and RT.

Test Input (Raw Title)	Regex Matches Stripped	Cleaned Output	Result Status
"Ebenezer: Early Access"	"Early Access", trailing ":"	"Ebenezer"	Clean String
"Ghost Soldier (Early Release)"	"Early Release", (), []	"Ghost Soldier"	Clean String
"Rocky 50th Anniversary Showing"	"Anniversary", "Showing"	"Rocky 50th"	Sanitized
"Avatar: The Way of Water - Dolby Cinema 3D"	"Dolby Cinema", "3D", trailing "-"	"Avatar: The Way of Water"	Clean String
"Special Screening: Dune Part Two (IMAX)"	"Special Screening", "IMAX", ()	"Dune Part Two"	Clean String

1.3 Review Text Length & Inline Link Truncation

Validates truncation boundaries for user review cards (max_len = 700 chars) to prevent embed body overflow.

Input Length	Condition	Formatted Output Pattern
650 chars	len <= 700	||[Full Text]|| [read full review](url)
700 chars	len == 700	||[Full Text]|| [read full review](url)

701 chars	len > 700	|||[Truncated Text...]||| [read more](url)

2. Truth Tables & Logic Decision Paths

2.1 Multi-Tier Rotten Tomatoes Candidate Fallback Logic

Evaluates candidate URL execution paths for standard releases, current-year releases, and legacy re-releases.

Movie Title	Current Year (year)	Original Year (orig_year)	Primary URL Tested (/m/slug)	Secondary Candidate (/m/slug_year)	Tertiary OMDb Fallback (Ratings [])	Final Output
Ebenezer	2026	2026	/m/ebenezer (Hit 200)	N/A	N/A	[38%](url)
Being Heumann	2026	2026	/m/being_heumann (Hit 200)	N/A	N/A	[86%](url)
Rocky	2026	1976	/m/rocky (404)	/m/rocky_2026 (404)	OMDb Array: 85%	[85%](url)
Unknown Indie	2026	2026	/m/unknown_indie (404)	/m/unknown_indie_2026 (404)	None Found (N/A)	N/A

2.2 Critic Score Normalization & Combined Average Math

Rules for calculating combined_average_10 across dynamic combinations of valid scores, TBD states, and missing values.

Rotten Tomatoes	IMDb	Metacritic	Valid Numeric Scores	Calculation	Displayed Combined Average
86% (8.6)	N/A	N/A	[8.6]	$\frac{8.6}{1} = 8.60$	**8.60 / 10.00**
38% (3.8)	5.6/10 (5.6)	[TBD]	[3.8, 5.6]	$\frac{3.8+5.6}{2} = 4.70$	**4.70 / 10.00**
N/A	7.9/10 (7.9)	98/100 (9.8)	[7.9, 9.8]	$\frac{7.9+9.8}{2} = 8.85$	**8.85 / 10.00**
95% (9.5)	8.1/10 (8.1)	70/100 (7.0)	[9.5, 8.1, 7.0]	$\frac{9.5+8.1+7.0}{3} = 8.$	**8.20 / 10.00**
N/A	[TBD]	[TBD]	None []	0.0	**0.00 / 10.00**

3. Regression & Feature Test Suite (v0.0.6)

Test ID	Module	Feature Description	Test Steps	Expected Result	Pass/Fail
RT-01	tasks.py	Async Ingestion Speed	Execute /force_refresh.	Total cycle completes in **< 5.0s**) with full showtime and metadata serialization.	[P]

RT-02	scraper.py	Multi-Tier RT Fallback	Query a title with a legacy release year (*Rocky*, 1976).	Scraper executes fallback loop, resolves score via OMDb or legacy slug, and displays valid percentage.	[P]
RT-03	scraper.py	Metacritic TBD Handling	Query an unreleased title (*Ebenezer*).	Embed card renders Metacritic field as **[TBD](url)** instead of generic N/A.	[P]
RT-04	scraper.py	Title Sanitization	Query movie with "Early Release", "Fan Event", or "Anniversary".	Tags are stripped cleanly; search query succeeds without trailing punctuation errors.	[P]
RT-05	cogs/user	/reviews Command Refactor	Execute /reviews [title] critic and /reviews [title] user.	Renders exact same visual embed layout, thumbnail, and sub-headers as /movie dropdown tabs.	[P]
RT-06	cogs/user	Combined Average Math	Verify average on movie with 1 valid score (*Being Heumann*).	Computes single-score average correctly (**8.60 / 10.00**) without division errors or N/A skewing.	[P]

RT-07	Ephemeral	Visibility Override	Execute /reviews movie_title:Ebenezer review_type:Critic visible_to_all:true.	Embed posts publicly to channel (ephemeral=False).	[P]

4. Test Suite Summary & Release Sign-Off

- **Total Test Cases Executed:** 8
- **Passed:** 8
- **Failed:** 0
- **Build Status: PASS (v0.0.6 Approved for Production Deployment)**